

FOOTPRINTING AND RECONNAISSANCE LAB REPORT

BACKGROUND	
Course:	CIS 170 - Hands-On Ethical Hacking and Network Defense
Term:	Spring 2026
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Submission Platform:	Canvas
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Lab Title:	Footprinting and Reconnaissance
Module Reference:	Footprinting and Reconnaissance
Lab Type:	Live Virtual Machine Lab
Date Completed:	2026-01-19
Time Spent:	Approximately 1 hour
ENVIRONMENT	
Platforms and Systems:	
– Primary environment: Practice Labs / Live Virtual Machine Labs controlled sandbox. – Primary systems available in the topology: PLABDC01 (Windows Server 2019 - Domain Server), PLABDM01 (Windows Server 2019 - Domain Member), PLABWIN10 (Windows 10 - Workstation), and PLABKALI01 (Kali 2019.2 - Linux Kali Workstation). – The work centered mainly on PLABKALI01 for Nmap, p0f, nslookup, theHarvester, and Maltego, with PLABWIN10 used for browser-based passive-reconnaissance tasks such as Whois, GHDB, temporary e-mail generation, and public-record searching. – Remote access to the virtual machines was provided through the course platform.	
Context and Constraints:	
– The module focused on reconnaissance methods including active scanning, passive information gathering, DNS querying, open-source intelligence, service detection, and graph-based analysis. – The practical work in this environment used Nmap, p0f, Whois services, Hunter, SHODAN, GHDB, nslookup, theHarvester, temp-mail, and Maltego. – Several passive-reconnaissance tasks relied on live public websites, so exact results could vary between runs. – Windows Updates were disabled in the Practice Labs environment, so some software behavior reflected a guided legacy setup rather than a current hardened baseline. – The module aligned most closely with exam objective 5.2 (Information Security Assessment Methodologies).	
Authorization and Scope:	
– All documented activity took place only inside approved academic systems and isolated practice-lab sandboxes or through public sources explicitly used by the lab module. – No unauthorized exploitation, persistence, or credential misuse was performed against third-party systems. – The actions documented here were performed for defensive understanding, attack-surface analysis, and ethical hacking education within the course environment. – Restricted material, live credentials, and personal data are omitted or redacted from this report.	
Work Objectives:	
– Review reconnaissance terminology, common tooling, and footprinting countermeasures. – Identify live hosts and perform discovery scans, port scanning, operating-system fingerprinting, and service probing in the lab network. – Use passive-reconnaissance sources such as Whois, Hunter, SHODAN, GHDB, DNS queries, theHarvester, temp-mail, Maltego, and public-record searching to collect external intelligence.	
Reconnaissance Terminology and Countermeasure Summary:	
– The lab distinguished active, passive, pseudonymous, Internet-based, and anonymous reconnaissance. – The need for footprinting was framed as understanding security posture, reducing attack area, collecting maximum information, and building a usable picture of the target network. – Key countermeasures included disabling unnecessary ports and services, firewall filtering, IDS monitoring, separate internal/external DNS, banner reduction, and user security awareness.	

TECHNICAL ANALYSIS

Active Reconnaissance with Nmap and p0f:

- On PLABKALI01, live-host identification and discovery scanning were performed with the following commands: `nmap -sP 192.168.0.0/24`, `nmap -sn 192.168.0.0/24`, `nmap -traceroute 192.168.0.0/24`, `nmap 192.168.0.1-4`, and `nmap 192.168.0.*`.
- These scans demonstrated subnet-wide host discovery, IP-range scanning, wildcard scanning, and route visibility to live systems within the lab environment.
- Alternative discovery approaches were then used with `nmap -PR 192.168.0.0/24`, `nmap -p 80 192.168.0.0/24`, and `nmap -sn -PS80 192.168.0.0/24` to compare ARP-based, port-based, and SYN-based identification methods.
- Port scanning was performed with `nmap 192.168.0.2`, `nmap -p 22 192.168.0.1`, `nmap -p 1-100 192.168.0.1`, `nmap -F 192.168.0.1`, `nmap -p- 192.168.0.1`, `nmap -sT 192.168.0.1`, `nmap -sU -p 53,80,3389 192.168.0.1`, and `nmap -top-ports 10 192.168.0.1`.
- These commands showed default scans, single-port checks, port-range scans, fast scans, full 65535-port scanning, TCP connect scans, UDP probing, and top-port analysis.
- Passive OS fingerprinting was performed with p0f while Firefox ESR generated intranet traffic, allowing passive operating-system and uptime-related fingerprinting from observed packets.
- Service probing and version detection were completed with `nmap 192.168.0.1` and `nmap -sV -O 192.168.0.1`, combining open-port identification with service-version and operating-system detection on the domain controller.

Passive Reconnaissance and OSINT Collection:

- On PLABWIN10, Microsoft Edge was used to browse <https://whois.com/whois/> and whois.domaintools.com, with www.practice-labs.com used as the search target on both services.
- These lookups exposed registrar information, whois data, IP history, server details, and domain registration context.
- On PLABKALI01, Firefox ESR was used to visit <https://hunter.io> and <https://shodan.io>. The Hunter search target was `google.com`, and the SHODAN search terms used were `VNC` and `webcamxp`.
- These tasks demonstrated public e-mail discovery, indexed devices, exposed services, and Internet-visible metadata.
- GHDB was accessed on PLABWIN10 through <http://www.exploit-db.com/google-hacking-database>, where the Files Containing Usernames category was reviewed and a result containing `xlsx`-style content was opened to demonstrate indexed sensitive file exposure.
- DNS querying on PLABKALI01 was performed with `nslookup practicelabs.com`, `nslookup -type=A practicelabs.com`, `nslookup -type=soa practicelabs.com`, `nslookup -type=A -debug practicelabs.com`, `nslookup -query=MX practicelabs.com`, `nslookup -type=ns practicelabs.com`, `nslookup practicelabs.com 192.168.0.1`, `nslookup -timeout=10 practicelabs.com`, and `nslookup -query=any practicelabs.com`.
- theHarvester was used with `theHarvester -d practice-labs.com -l 500 -b google` and `theHarvester -d theguardian.com -b bing` to gather e-mail, hostname, and source-derived intelligence.
- Temporary e-mail generation was demonstrated with <https://temp-mail.org>, and Maltego was used with a Domain entity renamed to `practice-labs.com` and the transforms `To DNS Name - MX` (mail server), `To Website [Quick lookup]`, `To IP Address [DNS]`, `To Server Technologies [Using BuiltWith]`, and `Machines > Footprint`.
- Public-record searching was demonstrated through <https://publicrecordreports.com>, which was the public-record search task used by the lab, with the search values `John Doe` and `Alabama`.

KEY COMMANDS, TOOLS, AND ARTIFACTS BY SYSTEM

PLABKALI01 — Active Recon / DNS / OSINT Tooling:

```
nmap -sP 192.168.0.0/24
nmap -sn 192.168.0.0/24
nmap -traceroute 192.168.0.0/24
nmap 192.168.0.1-4
nmap 192.168.0.*
nmap -PR 192.168.0.0/24
nmap -p 80 192.168.0.0/24
nmap -sn -PS80 192.168.0.0/24
nmap 192.168.0.2
nmap -p 22 192.168.0.1
nmap -p 1-100 192.168.0.1
nmap -F 192.168.0.1
nmap -p- 192.168.0.1
nmap -sT 192.168.0.1
nmap -sU -p 53,80,3389 192.168.0.1
nmap -top-ports 10 192.168.0.1
p0f
nmap -sV -O 192.168.0.1
nslookup practicelabs.com
nslookup -type=A practicelabs.com
nslookup -type=soa practicelabs.com
nslookup -type=A -debug practicelabs.com
nslookup -query=MX practicelabs.com
nslookup -type=ns practicelabs.com
nslookup practicelabs.com 192.168.0.1
nslookup -timeout=10 practicelabs.com
nslookup -query=any practicelabs.com
theHarvester -d practice-labs.com -l 500 -b google
theHarvester -d theguardian.com -b bing
```

PLABWIN10 — Browser-Based Passive Reconnaissance:

```
https://whois.com/whois/
www.practice-labs.com
whois.domaintools.com
www.practice-labs.com
https://hunter.io
google.com
https://shodan.io
VNC
webcamxp
http://www.exploit-db.com/google-hacking-database
https://temp-mail.org
https://publicrecordreports.com
John Doe
Alabama
```

Maltego Workflow and Primary Artifacts:

- Maltego was launched from Kali and used with a Domain entity renamed to practice-labs.com.
- The GUI transforms and machine actions used were To DNS Name - MX (mail server), To Website [Quick lookup], To IP Address [DNS], To Server Technologies [Using BuiltWith], and Machines > Footprint.
- Primary artifacts observed across the lab included Nmap scan outputs, p0f passive-fingerprint output, Whois result pages, SHODAN results, GHDB output, nslookup responses, theHarvester results, temp-mail page output, and the populated Maltego graph showing MX records, websites, IP addresses, detected server technologies, and a limited community-edition footprint graph.

DEFENSIVE MAPPING AND SUMMARY

Technique Summary:

- The completed work covered terminology review, host discovery, discovery scans, port scanning, passive operating-system fingerprinting, service probing, Whois analysis, DNS querying, browser-based OSINT, theHarvester, temp-mail, Maltego, and public-record searching.
- The recurring theme across the module was that effective reconnaissance combines direct network interaction with open-source public information.

MITRE ATT&CK Mapping:

- Reconnaissance: T1595 Active Scanning - Nmap host discovery, traceroute, and port scanning mapped directly to active collection from the target environment.
- Reconnaissance: T1590 Gather Victim Network Information - subnet discovery, DNS analysis, service exposure, and route visibility aligned with victim-network intelligence gathering.
- Reconnaissance: T1596 Search Open Websites/Domains - Whois, Hunter, SHODAN, GHDB, and public record searching mapped to open-source intelligence gathering from public services.

Detection Opportunities:

- Repeated sweeps, scan attempts, version checks, and service probes are detectable through firewall logging, IDS/IPS telemetry, and host monitoring.
- Public exposure through banners, weak DNS hygiene, indexed files, and excessive metadata creates reconnaissance opportunities before exploitation begins.
- Search-engine discoverability of usernames, files, devices, and domain information expands the attack surface significantly.

Defensive Controls:

- Disable unnecessary ports and services and minimize exposed banners wherever possible.
- Use firewalls, IDS/IPS, and routine monitoring to detect footprinting and scanning patterns.
- Use separate internal and external DNS where practical and harden public DNS records.
- Educate users against oversharing on public platforms and social media.
- Perform external footprinting against your own organization regularly to understand what is publicly visible to reconnaissance tools and search/indexing services.

Completion Verification:

- Work was completed across conceptual reconnaissance review, active Nmap scanning, passive p0f fingerprinting, DNS querying, theHarvester execution, browser-based OSINT collection, and Maltego graph analysis.
- Completion was evidenced by live-host scan outputs, port-state results, p0f fingerprints, Whois results, SHODAN outputs, GHDB output, DNS query responses, theHarvester results, temp-mail display, and populated Maltego entities.
- The actual commands and search targets used in the lab are preserved directly in this report for accurate documentation and reproducibility.

Key Findings:

- 1) Reconnaissance is most effective when direct network scanning and passive OSINT are combined.
- 2) Nmap provided broad visibility into host availability, route behavior, open ports, and exposed services within the subnet.
- 3) Passive sources such as Whois, SHODAN, GHDB, and theHarvester showed how much intelligence can be gathered before exploitation starts.
- 4) DNS records, indexed documents, public metadata, and detected technologies materially expand the attack surface.
- 5) The strongest defensive takeaway was that organizations must continuously assess what they expose to scanners and public indexing services.

Issues / Deviations / Notes:

- Several passive-reconnaissance tasks relied on live public services, so exact results could vary from the walkthrough and screenshots.
- Some tools depended on account creation or community-edition limitations, especially SHODAN and Maltego.
- Certain outputs, such as hostname resolution in theHarvester, were limited by the internal lab environment and firewall behavior.
- The lab text referenced AnyWho conceptually, but the actual public-record search task used publicrecordreports.com.

Reflection:

- This lab tied together host discovery, port and service analysis, passive OS fingerprinting, DNS interrogation, and public-source intelligence gathering into a single workflow.
- The strongest takeaway was that meaningful target intelligence can be built long before any exploit attempt occurs, which makes reconnaissance visibility a critical defensive concern.